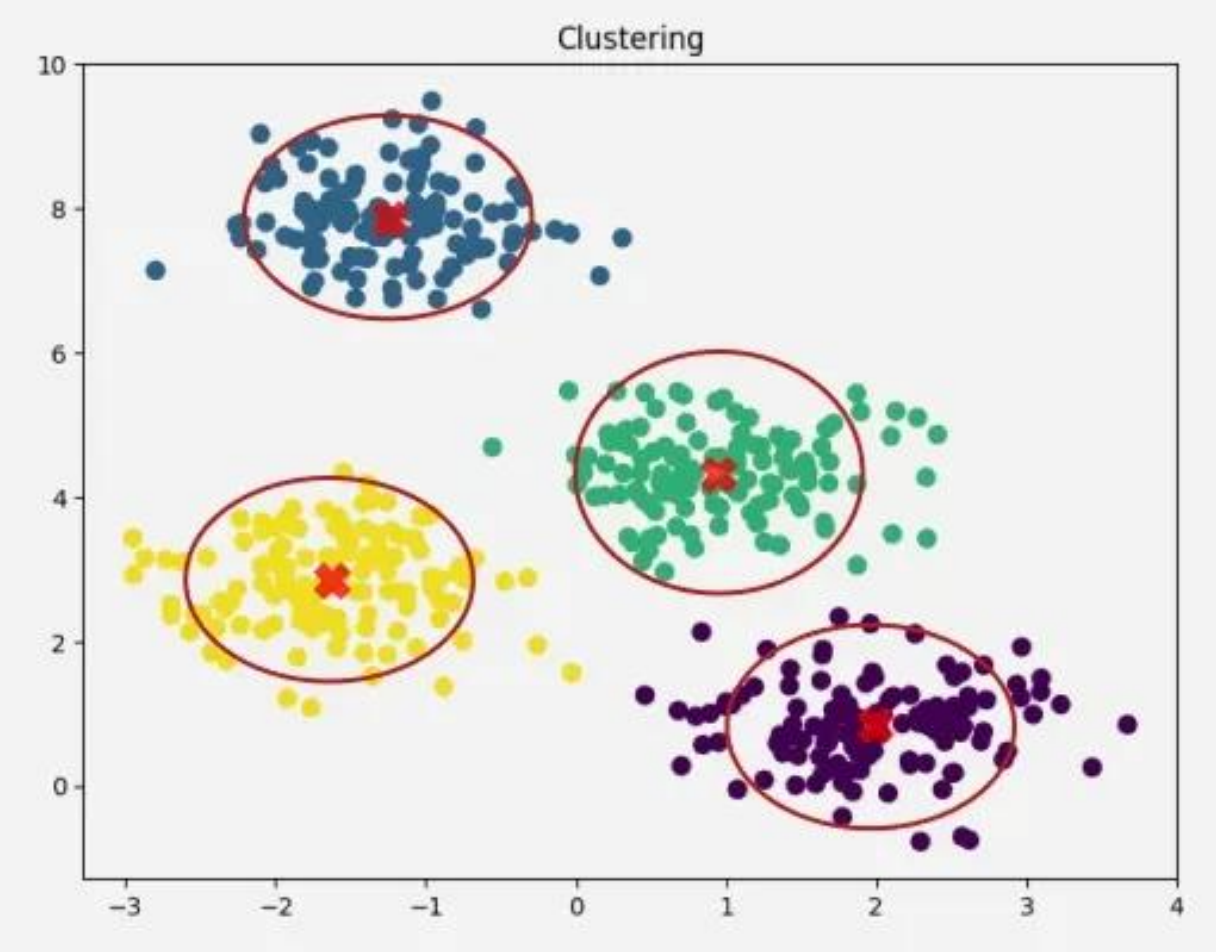
Data Mining

Cluster Analysis

Clustering

- A data mining and machine learning technique used to group a set of objects or data points into clusters (groups) such that:
  - Objects within the same cluster are more similar to each other (based on a similarity or distance measure), and
  - Objects in different clusters are dissimilar from one another.
- Given a dataset $D = \{x_1, x_2, \ldots x_n\}$,clustering aims to partition it into k clusters $C_1, C_2, \ldots, C_k$such that:
  - $\bigcup_{i=1}^{k} C_i = D$ *and* $C_i \cap C_j = \emptyset$ *for* $i \neq j$
  - and similarity within each $C_i$is maximized, while similarity between clusters is minimized.

Clustering



Goal of Clustering

- Discover natural groupings or patterns in data
- Reduce data complexity by summarizing large datasets
- Support further analysis like classification, outlier detection, or data visualization

Clustering Techniques

- Partitioning Methods: divide data into k groups
  - k-Means
  - k-Medoids
- Hierarchical Methods: create cluster trees
  - AGNES
  - DIANA
- Density-Based Methods: identify dense regions
  - DBSCAN

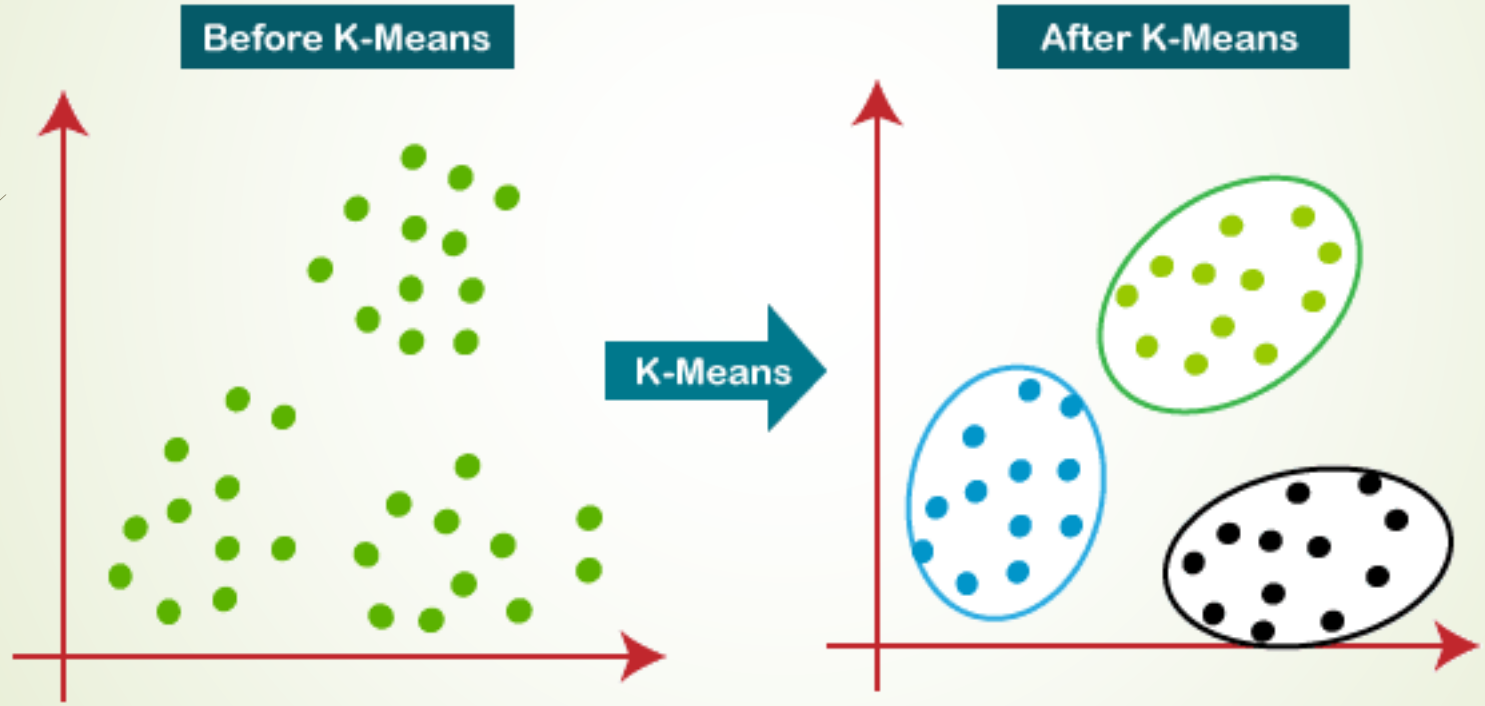
Requirements for Cluster Analysis

- Scalability: must handle large datasets efficiently
- Ability to handle different shapes: arbitrary-shaped clusters
- High-dimensionality handling: relevant feature selection
- Interpretability: clusters must be meaningful and usable
- Robustness: tolerant to noise and outliers
- Flexibility: discover clusters of different sizes and densities

k-Means

- K-means clustering is an unsupervised machine learning algorithm that groups similar data points into a pre-defined number of clusters, denoted by k.
- **Initialization**: The algorithm starts by randomly choosing k initial centroids, which are the centre points of the clusters.
- **Assignment**: Each data point is assigned to the cluster whose centroid is the nearest, typically using a distance metric like the Euclidean distance.
- **Update**: The algorithm recalculates the position of each of the k centroids. The new centroid is the mean of all the data points that were assigned to that cluster.
- **Iteration**: The assignment and update steps are repeated until the centroids stabilize, meaning the cluster assignments no longer change and the algorithm has converged.

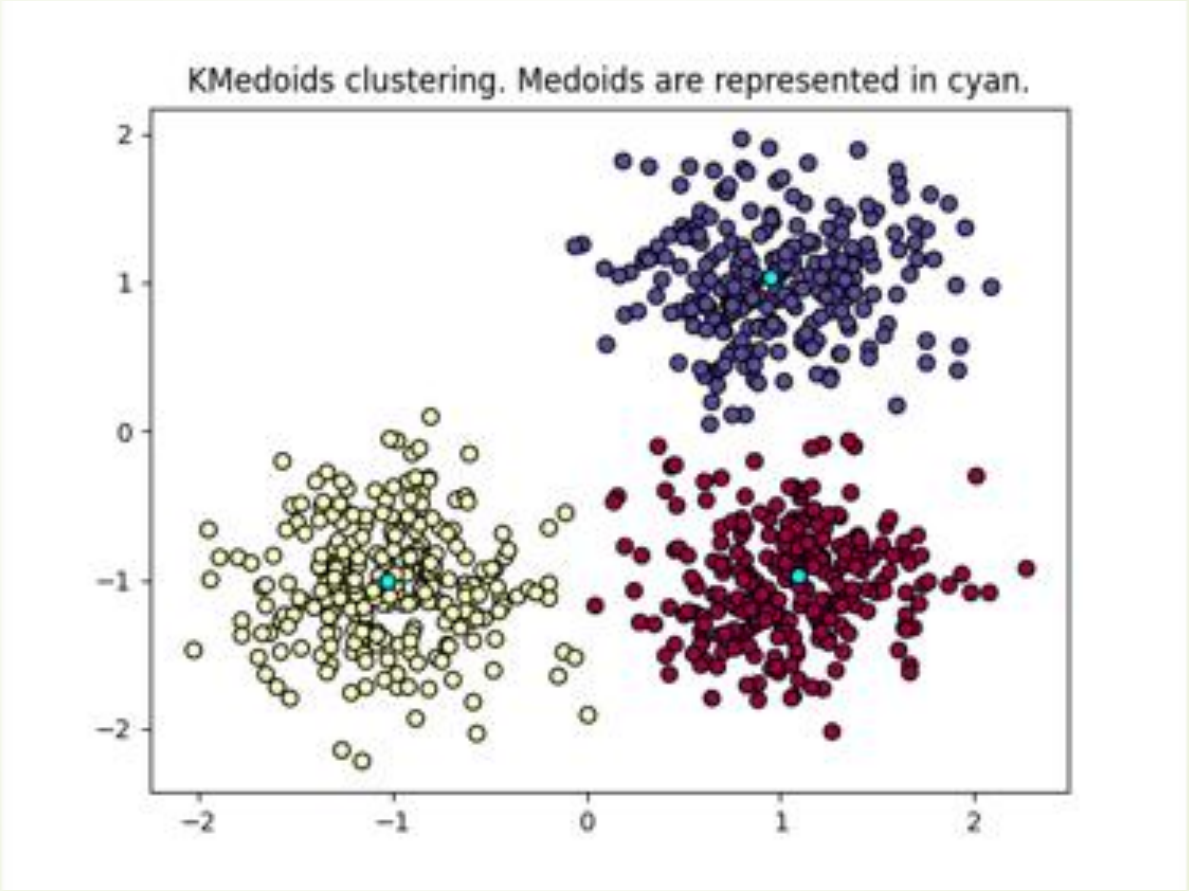
k-Means



k-Medoids

- An unsupervised learning algorithm that partitions a dataset into K clusters, using actual data points called medoids as cluster centres instead of the mean used in K-Means.
- **Initialization**: Select K data points from the dataset to serve as initial medoids.
- **Assignment**: Assign each data point to the cluster whose medoid is closest, using a distance metric like Euclidean or Manhattan distance.
- **Update (Swapping)**: For each cluster, consider swapping the current medoid with a non-medoid data point.
- **Cost Calculation**: Calculate the total cost (sum of distances to the medoid) for each possible swap.
- **Selection**: If a swap results in a lower total cost, the new medoid becomes permanent.
- **Iteration**: Repeat the assignment and swapping steps until no further changes in medoids lead to a reduction in cost, indicating the algorithm has converged.

k-Medoids


KMedoids clustering. Medoids are represented in cyan.
2
1
0
−1
−2
−2
−1
0
1
2

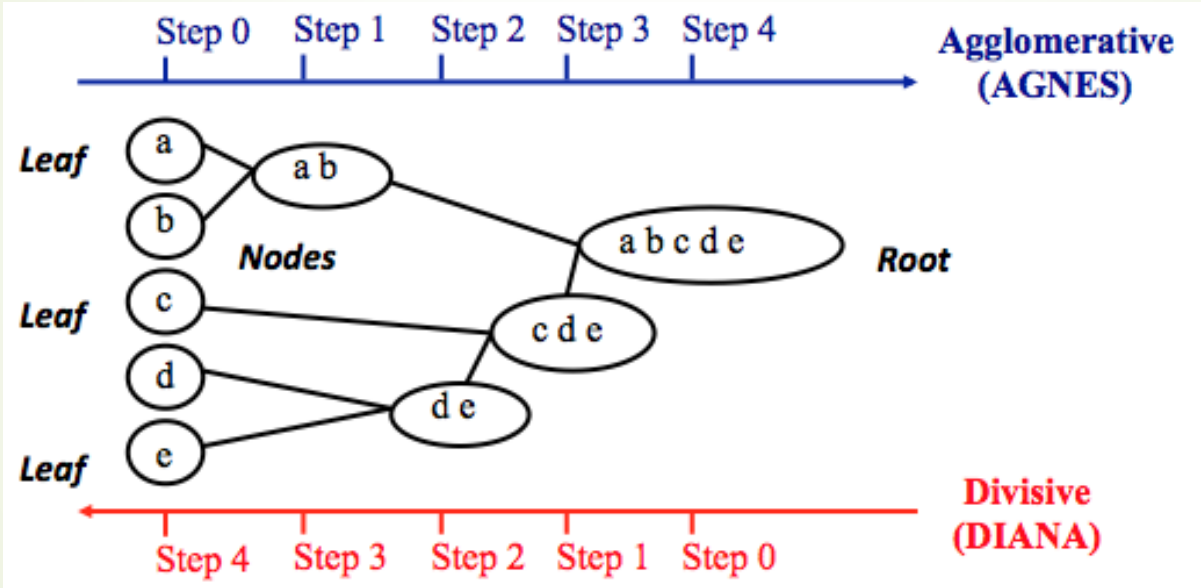
Hierarchical Clustering

- An unsupervised machine learning algorithm that groups data points into a tree-like hierarchy of clusters.
- **Agglomerative (bottom-up) AGNES:**
  - This is the most common approach.
  - It starts by treating each data point as its own cluster.
  - It then iteratively merges the two closest clusters until all data points are in a single, large cluster.

Hierarchical Clustering

- **Divisive (top-down) DIANA :**
  - This approach works in the opposite way.
  - It starts with all data points in one cluster.
  - It then progressively splits the cluster into smaller and smaller sub-clusters until each data point is in its own cluster.

Hierarchical Clustering



Hierarchical Clustering

- **BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)**
  - Designed for very large datasets
  - Builds a CF (Clustering Feature) Tree summarizing data
  - Incremental and memory-efficient
  - Handles noise well

Density-Based Method – DBSCAN

- Groups points that are densely packed
- Parameters:
  - ε (eps): neighborhood radius
  - MinPts: minimum points to form dense region
- The algorithm starts with an unvisited point and finds its neighbors within the ε-neighborhood. If there are at least MinPts neighbors, a new cluster is formed. The cluster expands by adding neighbors of core points until no more points can be added.

Density-Based Method – DBSCAN

